

Supplementary material for “Lightweight Large Language Model-Driven Semantic Dispatch Framework for Electric Vehicle Virtual Power Plants”

I. EV-LEVEL FLEXIBILITY ENVELOPE CONSTRUCTION

For charging station s , the EV-level upper and lower flexibility boundaries used in the main paper are constructed as

$$p_{s,i,\kappa}^+ = \begin{cases} p_{s,i}^{\max}, & a_{s,i} \leq \kappa \leq d_{s,i}, \\ 0, & \kappa < a_{s,i} \text{ or } \kappa > d_{s,i}, \end{cases} \quad (1a)$$

$$p_{s,i,\kappa}^- = \begin{cases} -v_{s,i} p_{s,i}^{\max}, & a_{s,i} \leq \kappa < d_{s,i}, \\ 0, & \kappa < a_{s,i} \text{ or } \kappa \geq d_{s,i}, \end{cases} \quad (1b)$$

$$e_{s,i,\kappa}^+ = \begin{cases} 0, & \kappa \leq a_{s,i}, \\ \min \left\{ e_{s,i,\kappa-1}^+ + \Delta T^\kappa \eta_{s,i} p_{s,i,\kappa}^+, e_{s,i}^{\text{tar}} \right\}, & a_{s,i} < \kappa < d_{s,i}, \\ e_{s,i}^{\text{tar}}, & \kappa \geq d_{s,i}, \end{cases} \quad (1c)$$

$$e_{s,i,\kappa}^- = \begin{cases} 0, & \kappa \leq a_{s,i}, \\ \max \left\{ e_{s,i,\kappa-1}^- + \Delta T^\kappa \frac{p_{s,i,\kappa}^-}{\eta_{s,i}}, 0 \right\}, & a_{s,i} < \kappa \leq \kappa_{s,i}^L, \\ \min \left\{ e_{s,i,\kappa-1}^- + \Delta T^\kappa \eta_{s,i} p_{s,i,\kappa}^+, e_{s,i}^{\text{tar}} \right\}, & \kappa_{s,i}^L < \kappa < d_{s,i}, \\ e_{s,i}^{\text{tar}}, & \kappa \geq d_{s,i}, \end{cases} \quad (1d)$$

$$\kappa_{s,i}^L = \max \left\{ a_{s,i}, d_{s,i} - \left\lceil \frac{e_{s,i}^{\text{tar}}}{\Delta T^\kappa \eta_{s,i} p_{s,i}^{\max}} \right\rceil \right\}. \quad (1e)$$

where $\kappa_{s,i}^L$ is the latest interval at which EV (s, i) can stay on the lower energy envelope while still reaching the departure target.

II. INTRA-DAY OBJECTIVE COST DEFINITIONS

The compact intra-day objective in the main paper consists of four cost terms:

$$C_{\text{ene}}^{\text{RT}} = \sum_{\tau \in \mathcal{T}^{\text{RT}}} \hat{\lambda}_\tau^{\text{RT}} (P_\tau^{\text{VPP,RT}} - P_\tau^{\text{VPP,DA}}) \Delta T^{\text{RT}}, \quad (2a)$$

$$C_{\text{dev}}^{\text{RT}} = \sum_{\tau \in \mathcal{T}^{\text{RT}}} c^{\text{dev}} |P_\tau^{\text{VPP,RT}} - P_\tau^{\text{VPP,DA}}| \Delta T^{\text{RT}}, \quad (2b)$$

$$C_{\text{op}}^{\text{RT}} = \sum_{\tau \in \mathcal{T}^{\text{RT}}} \sum_{s \in \mathcal{S}} c^{\text{op}} |P_{s,\tau}^{\text{CS}}| \Delta T^{\text{RT}}, \quad (2c)$$

$$C_{\text{env}}^{\text{RT}} = c^{\text{env}} \sum_{\tau \in \mathcal{T}^{\text{RT}}} \sum_{s \in \mathcal{S}} \zeta_{s,\tau}. \quad (2d)$$

where $C_{\text{ene}}^{\text{RT}}$ is the intra-day energy-settlement cost, $C_{\text{dev}}^{\text{RT}}$ penalizes deviation from the mapped day-ahead dispatch trajectory, $C_{\text{op}}^{\text{RT}}$ represents the CS operation cost, and $C_{\text{env}}^{\text{RT}}$ penalizes relaxation of the lower energy envelope.

III. EV-LEVEL DISAGGREGATION MODEL

After the station-level dispatch power is obtained, the EV-level disaggregation model allocates the station reference to individual EV charging and discharging trajectories. For CS s , let $P_{s,\tau}^{\text{ref}}$ denote the station-level reference power. The disaggregation problem is formulated as:

$$\min \sum_{\tau \in \mathcal{T}^{\text{RT}}} (P_{s,\tau}^{\text{CS}} - P_{s,\tau}^{\text{ref}})^2 + c^{\text{sf}} \sum_{i \in \mathcal{N}_s} \sigma_{s,i} \quad (3a)$$

$$P_{s,\tau}^{\text{CS}} = \sum_{i \in \mathcal{N}_s} p_{s,i,\tau}^{\text{ev}} \quad (3b)$$

$$p_{s,i,\tau}^{\text{ev}} = p_{s,i,\tau}^{\text{ch}} - p_{s,i,\tau}^{\text{dis}} \quad (3c)$$

$$e_{s,i,\tau+1} = e_{s,i,\tau} + \Delta T^{\text{RT}} \eta_{s,i} p_{s,i,\tau}^{\text{ch}} - \Delta T^{\text{RT}} \frac{p_{s,i,\tau}^{\text{dis}}}{\eta_{s,i}} \quad (3d)$$

$$0 \leq p_{s,i,\tau}^{\text{ch}} \leq p_{s,i}^{\text{max}} \delta_{s,i,\tau}^{\text{ch}}, \quad 0 \leq p_{s,i,\tau}^{\text{dis}} \leq v_{s,i} p_{s,i}^{\text{max}} \delta_{s,i,\tau}^{\text{dis}} \quad (3e)$$

$$\delta_{s,i,\tau}^{\text{ch}} + \delta_{s,i,\tau}^{\text{dis}} \leq 1, \quad \delta_{s,i,\tau}^{\text{ch}}, \delta_{s,i,\tau}^{\text{dis}} \in \{0, 1\} \quad (3f)$$

$$\begin{cases} e_{s,i,\tau} = 0, & \tau \leq a_{s,i} \\ 0 \leq e_{s,i,\tau} \leq e_{s,i}^{\text{tar}}, & a_{s,i} < \tau < d_{s,i}, \quad \sigma_{s,i} \geq 0 \\ e_{s,i,\tau} + \sigma_{s,i} = e_{s,i}^{\text{tar}}, & \tau \geq d_{s,i} \end{cases} \quad (3g)$$

where $\sigma_{s,i}$ is the departure-energy shortfall of EV (s, i) , and c^{sf} is the corresponding penalty coefficient. The first objective term tracks the station-level reference power, while the second term penalizes unmet departure energy.

IV. KEY LLM HYPERPARAMETER SETTINGS

This supplementary file reports the main LLM hyperparameters used in the semantic parsing experiments. Only the settings that materially affect parser training and inference are included here.

A. Supervised Fine-Tuning

The first-stage supervised fine-tuning adapts the lightweight model to the dispatch-oriented semantic event schema and the required structured JSON output protocol.

TABLE I
MAIN HYPERPARAMETERS FOR LoRA-SFT

Item	Setting
Base model	Qwen2.5-0.5B-Instruct
Training samples	600
Fine-tuning method	LoRA-SFT
LoRA rank / alpha / dropout	16 / 32 / 0
Maximum sequence length	4096 tokens
Learning rate	5.0×10^{-5}
Scheduler	Cosine
Optimizer	AdamW
Epochs	3
Per-device batch size	2
Gradient accumulation steps	8
Numerical precision	bfloat16

B. Reinforcement Fine-Tuning

The second-stage reinforcement fine-tuning is initialized from the LoRA-SFT model. The compared GRPO-family methods use the same policy-training hyperparameters, while differing in reward construction.

TABLE II
MAIN HYPERPARAMETERS FOR GRPO-FAMILY FINE-TUNING

Item	Setting
Initialization	LoRA-SFT model
Training samples	600
Fine-tuning method	LoRA-based GRPO-family update
LoRA rank / alpha / dropout	16 / 32 / 0.1
KL coefficient	0.04
Sampling temperature during training	0.9
Learning rate	3.0×10^{-5}
Scheduler	Cosine
Optimizer	AdamW
Weight decay	0.1
Warmup ratio	0.1
Epochs	15
Per-device batch size	2
Gradient accumulation steps	4
Number of sampled generations	4
Maximum prompt length	4096 tokens
Maximum completion length	1024 tokens
Maximum gradient norm	0.1
Numerical precision	bfloat16

TABLE III
REWARD-SPECIFIC SETTINGS

Method	Reward setting	Buffer size	Update rate	Bonus
GRPO	Fixed parsing reward and format-boundary reward	–	–	–
LoRA-GRPO	Fixed parsing reward and format-boundary reward	–	–	–
LoRA-SGRPO	Static error-aware reward over four parsing error types	300	0	0.25
LoRA-AGRPO	Adaptive error-aware reward over four parsing error types	300	0.02	0.25

C. Parser Inference

The complete semantic dispatch scheme uses the LoRA-AGRPO parser for online conversion from natural-language requests to structured semantic events.

TABLE IV
MAIN HYPERPARAMETERS FOR PARSER INFERENCE

Item	Setting
Parser model	Qwen2.5-0.5B LoRA-AGRPO parser
Test samples	300
Maximum new tokens	2048
Top- p	0.7
Temperature	0.95
Device	Single GPU
Output protocol	Boundary-delimited JSON

D. Token Accounting and Marginal Response Cost

For the deployment-oriented comparison in the case study, the marginal response cost of cloud-based LLMs is estimated from the measured token consumption during benchmark testing. The input side includes the system prompt and the user request, while the output side corresponds to the generated assistant response. The resulting average token counts are summarized in Table V.

TABLE V
AVERAGE TOKEN CONSUMPTION PER USER REQUEST

Model	Samples	Avg. input tokens	Avg. output tokens
Qwen2.5-0.5B LoRA-AGRPO parser	300	747.66	88.50
Qwen3.6-35B-A3B	300	784.65	1291.39
ChatGPT-5.5	300	707.26	155.85

Let $\bar{n}_{\text{in}}$ and $\bar{n}_{\text{out}}$ denote the average input and output token counts per request, respectively. Given the input and output prices p_{in} and p_{out} per one million tokens, the marginal cost per response is computed as

$$C_{\text{resp}} = \frac{\bar{n}_{\text{in}}}{10^6} p_{\text{in}} + \frac{\bar{n}_{\text{out}}}{10^6} p_{\text{out}}. \quad (4)$$

For models priced in CNY, the result is converted to USD using the exchange rate 1 USD = 6.8248 CNY. Table VI reports the corresponding cost estimates. These values only represent API-style marginal calling costs and do not include local hardware depreciation or electricity cost. The local parser is therefore assigned zero external API calling cost in the main case-study table.

TABLE VI
ESTIMATED MARGINAL COST PER RESPONSE FOR CLOUD-BASED LLMs

Model	Input price	Output price	Currency	Cost (USD)
Qwen3.6-35B-A3B	1.8 / 1M tokens	10.8 / 1M tokens	CNY	0.0023
ChatGPT-5.5	5 / 1M tokens	30 / 1M tokens	USD	0.0082